

THE TRUE ECONOMICS OF OPTIMISATION

The Economics of Autonomous HVAC Optimisation

Increasing engineering capacity without increasing engineering headcount

The question is no longer simply how much energy optimisation can save. It is how much engineering resource is required to deliver and sustain those savings.

Most HVAC optimisation business cases focus on the value of the energy saved. Far less attention is given to the cost of the people required to create and preserve those savings. For large estates, that missing cost can materially change the economics.

WHAT IS MISSING FROM THE OPTIMISATION BUSINESS CASE?

1 Identify

Find the opportunity in the data.

2 Deliver

Implement the change through the BMS.

3 Sustain

Keep retuning as conditions change.

A BETTER COMMERCIAL EQUATION

Apparent ROI

Energy savings - software or project cost

True optimisation economics

Verified savings - solution cost - engineering resource

**It is not engineering replacement.
It is engineering leverage.**

What does it cost to deliver the saving?

A predicted energy opportunity is not the same thing as a realised saving. Recommendations still have to be investigated, implemented, checked and revisited as weather, occupancy and building behaviour change.

THE ENGINEERING WORK BEHIND THE HEADLINE NUMBER

Review the data

Trend analysis, schedules, setpoints and plant behaviour.

Investigate recommendations

Determine what should actually change and whether it is safe.

Implement changes

Modify BMS strategies, schedules or selected setpoints.

Verify the result

Check that the intervention delivered the expected outcome.

Revisit the building

Respond as weather, occupancy and system behaviour change.

Repeat across the estate

Scale the same process across many buildings.

That time has a cost whether it is provided by an internal BMS team, an FM provider, a controls contractor or an external optimisation consultant. Yet it is frequently absent from the headline ROI.

AUTONOMOUS OPTIMISATION CHANGES THE EQUATION

Analytics-led optimisation

Software identifies opportunities. Engineers still need to interpret, implement, verify and revisit the changes.

Insight + engineering effort

Autonomous optimisation

myCoreAI continuously adjusts selected HVAC parameters through the existing BMS every 15 minutes within agreed limits.

Optimisation built into the service

The cost of continuous optimisation is built into the solution rather than hidden in engineering resource.

WHY THIS MATTERS AT ESTATE SCALE

More buildings normally mean more trends to review, more interventions to make and more performance to monitor. Eventually, sustaining optimisation becomes a resource problem as much as an energy problem.



Engineering leverage, not engineering replacement

BMS engineers remain essential. Autonomous HVAC optimisation does not remove the need for controls expertise; it changes how that expertise is used.

Reduce energy

Continuous adjustment reduces avoidable HVAC demand.

Increase capacity

Optimisation scales without a proportional increase in engineering headcount.

Protect expertise

Engineers concentrate on faults, upgrades, commissioning and complex decisions.

EVIDENCE IN OPERATION: UNIVERSITY OF ST ANDREWS

At the Scottish Oceans Institute, myCoreAI was introduced through the existing BMS while the university retained its own BMS engineering capability. Over the first 12 months, the project delivered:

158 MWh

verified energy reduction

21%

HVAC electricity reduction

9%

heat reduction

29.3 tCO₂e

carbon reduction

The university still needs BMS engineers. That is precisely the point. Autonomous optimisation means their expertise does not have to be consumed by continuous routine retuning.

HOW OPTIMISATION SHOULD NOW BE EVALUATED

- How much energy can the solution demonstrably save?
- How much ongoing engineering input is required to deliver those savings?
- Does optimisation continue automatically as conditions change?
- Can the approach scale without proportionally scaling headcount?
- Are the savings verified after the full cost of delivery is considered?

The cost of continuous optimisation is built into the solution rather than hidden in engineering resource.